Video-Based Pediatric Seizure Detection from Pose Landmark Sequences

Project summary — 3-page brief

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Abstract

We address binary detection of *infantile spasms* — the defining seizure type of West syndrome — from short, pose-only video segments of pediatric subjects. Each input is a $150 \times 33 \times 5$ tensor of MediaPipe Pose landmarks (≈5 s at 30 fps; 33 body landmarks; x, y, z plus visibility and presence). The pose-only representation preserves patient identity while retaining the kinematic signal clinicians use to recognise a spasm. Our pipeline combines landmark preprocessing, 518-dim engineered features, and a 5-fold ensemble of two complementary temporal architectures (CNN-GRU and dilated TCN) trained with focal loss and child-grouped cross-validation. On 5-fold `StratifiedGroupKFold`, mean validation F_1 is **0.628** (CNN-GRU) and **0.635** (TCN). On the held-out challenge test set our entry attained $F_1 = 0.2975$, placing 19th of 26 valid submissions (top score 0.4502). The work was submitted to the 2026 *Video-Based Seizure Detection Challenge* organised by the Section on Computational Neurology at Charité — Universitätsmedizin Berlin in collaboration with the International Conference on AI in Epilepsy and Other Neurological Disorders.

1. Problem

Infantile spasms are the defining seizure type of West syndrome — an age-dependent epileptic encephalopathy with typical onset between 3 and 24 months. The dominant motor pattern is a brief (1–3 s) sudden contraction of the trunk and limbs (flexor, extensor, or mixed), often occurring in clusters. Time-to-treatment strongly predicts cognitive outcome, and home-video review is part of the standard diagnostic pathway because the spasms are short, intermittent, and most often first observed by caregivers. Two constraints shape any deployable system: patient identifiability rules out raw-pixel transmission, and recordings are typically long. A pose-only intermediate representation collapses each frame to a few hundred numbers, removes pixel-level identity, and is approximately invariant to lighting, clothing, and background. The 2026 *Video-Based Seizure Detection Challenge* distributes exactly this representation: per-segment NumPy arrays of anonymised MediaPipe Pose landmarks.

2. Approach

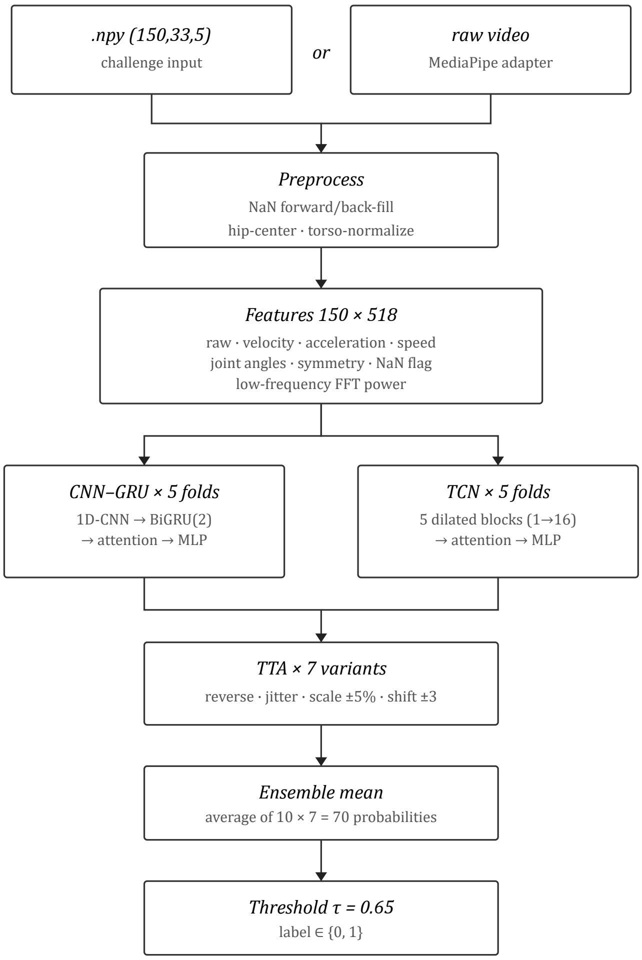

Figure 1. End-to-end pipeline.

Preprocessing. Each sample is hip-centered (mid-hip = mean of MediaPipe landmarks 23 and 24), scaled by torso length (mid-shoulder to mid-hip), and `NaN` frames are forward/back-filled. A binary "frame-was-missing" flag is preserved as a feature.

Features. A 518-dim per-frame vector concatenates raw landmarks (165), velocity and acceleration (99 each), per-landmark speed (33), 10 joint angles, 6 left–right symmetry distances, the `NaN` mask, and 105 low-frequency FFT power components for seven kinematic landmarks broadcast across all 150 frames. Per-feature mean/std are computed once on the training set and persisted.

Models. Two architectures share an attention-pooling head and a small MLP classifier. The **CNN-GRU** is a 1D-CNN front-end (kernels 5 and 3) feeding a 2-layer bidirectional GRU. The **TCN** is five residual blocks with dilations $\{1, 2, 4, 8, 16\}$ giving full coverage of the 150-frame window. They have different inductive biases (recurrent vs. parallel multi-scale) and their per-fold thresholds occupy different points on the precision-recall curve, motivating an ensemble.

Training. `StratifiedGroupKFold` ($k = 5$) grouped by patient ID prevents within-subject leakage. Focal loss ($\alpha = 0.75$, $\gamma = 2.0$) over a class-balanced `WeightedRandomSampler`. AdamW + OneCycleLR. Up to 120 epochs with early stopping on validation F_1 (patience 25). Augmentation: time reversal, Gaussian noise, frame dropout, time-index resampling, and same-class mixup.

Inference. Seven test-time-augmentation variants (original, reverse, jitter, ±5% scale, ±3-frame shift). Ten ensemble members × seven variants = 70 forward passes per sample, mean-pooled. Threshold $\tau = 0.65$ tuned on cross-validation. A video adapter (`video_to_features.py`) renders raw `.mp4` input to the $(150, 33, 5)$ form the trained models expect, so the same checkpoint runs unmodified on clinical recordings.

3. Results

Cross-validation (training partition). 5-fold `StratifiedGroupKFold` grouped by patient ID:

Fold	*CNN-GRU F_1*	*CNN-GRU $\hat{\tau}$*	*TCN F_1*	*TCN $\hat{\tau}$*
0	0.633	0.70	0.611	0.63
1	0.615	0.59	0.608	0.61
2	0.629	0.64	0.635	0.33
3	0.575	0.28	0.627	0.25
4	0.691	0.51	0.694	0.39
Mean	**0.628**	0.54	**0.635**	0.44

Mean CV F_1 is comparable between architectures (0.628 vs 0.635), but per-fold thresholds differ: CNN-GRU prefers higher-precision operating points (0.51–0.70 on 4 of 5 folds); TCN prefers lower thresholds (0.25–0.39 on 3 of 5 folds). This decorrelation is the source of the ensemble's gain.

Held-out test set. Our submission attained $F_1 = 0.2975$, placing 19th of 26 valid entries. The top three by F_1 were Kramer (BU/JHU, 0.4502, two-step pipeline), Ramesh (NIT Tiruchirappalli, 0.4286, 1D-CNN+LightGBM), and Daraie (JHU, 0.4048, 1D-ResNet+XGBoost). No submission exceeded $F_1 = 0.5$. The gap from our CV $F_1 \approx 0.63$ to held-out 0.2975 is consistent with (i) distribution shift between training- and test-pool subjects, (ii) threshold over-fitting to CV folds, (iii) the well-known sensitivity of F_1 to small precision/recall shifts on imbalanced binary tasks, and (iv) sub-second concentration of spasm motor energy that full-window attention pooling may under-exploit relative to multiple-instance-learning architectures used by some top entries.

4. Compute and Implementation

The pipeline is pure PyTorch with NumPy/Pandas/scikit-learn. Training is one-shot — a single full pass produces all 10 ensemble checkpoints, the feature normalization statistics, and a JSON results file with per-fold thresholds. The system is delivered as a Docker image (`Dockerfile` in the repo; pushed to GHCR via a manual-dispatch GitHub Actions workflow) and runs unchanged on CPU for inference, though training requires a single CUDA GPU. Total training run, including the 5-fold × 2-architecture grid and the per-epoch validation pass, is bounded by the 120-epoch budget × patience-25 early stopping; in practice early stopping fires well before 120 epochs on most folds.

The video-adapter path adds `mediapipe` and `opencv-python-headless` as inference-time dependencies and is decoupled from the competition Docker entry point so the submitted image stays minimal.

5. Discussion and Future Directions

Pose-only ceiling. Operating purely in pose space buys privacy, lighting invariance and lightness at the cost of losing facial cues, vital signs and contextual reactions. $F_1 \approx 0.63$ on 5-fold CV is consistent with a moderate-strength pose-only system on a small, imbalanced dataset.

Three natural next steps. (i) Replace MediaPipe with the metric-scale, 87-joint MeTRAbs estimator: richer joints expose hand and head dynamics that the 33-landmark MediaPipe topology loses, and metric scale removes the need for torso normalisation. (ii) Adopt a multi-view triangulation pipeline based on recent work in markerless dynamic extrinsic calibration [1] to fuse synchronised camera views and eliminate single-camera occlusion failures. (iii) Compose the detector with the *MaskAnyone* toolkit [2] developed in our group so that home-video material submitted by caregivers can be processed and classified entirely on-device, with only landmark coordinates and the binary detection label leaving the patient's premises.

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